

Advanced Satellite For TELEMETRY AND REMOTE ACCESS

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Advanced Satellite for Telemetry and Remote Access (ASTRA) is a CubeSat model designed to offer hands-on experience in satellite technology. It serves as a practical platform for learning and innovation in space systems.

While conventional education often leans heavily on theory, it can leave students underprepared for real world engineering challenges. ASTRA addresses this by allowing learners to apply classroom concepts through the design, construction, and operation of a working satellite model. The result is meaningful exposure to satellite systems and embedded technologies. The required components are listed in the Bill of Materials table.

Circuit and working

Fig. 1 shows the circuit diagram of the CubeSat model. The setup is primarily based on the ESP32 development board, the MPU-6050 sensor module, the NEO-6M GPS module, the INA219 current sensor, and supporting components.

An onboard computer (OBC) is included to manage the satellite's power system and to transmit real-time data, such as current, voltage, load, and power consumption, via a Wi-Fi network. The source code can be modified to transmit and receive additional payload data, including UV detection, GPS coordinates, and IMU sensor readings.

In addition to the current sensor, the model includes an MPU6050 gyro sensor for orientation and a GPS module for location data. However, these data are not currently transmitted to the hosted webpage. Test code is provided for these

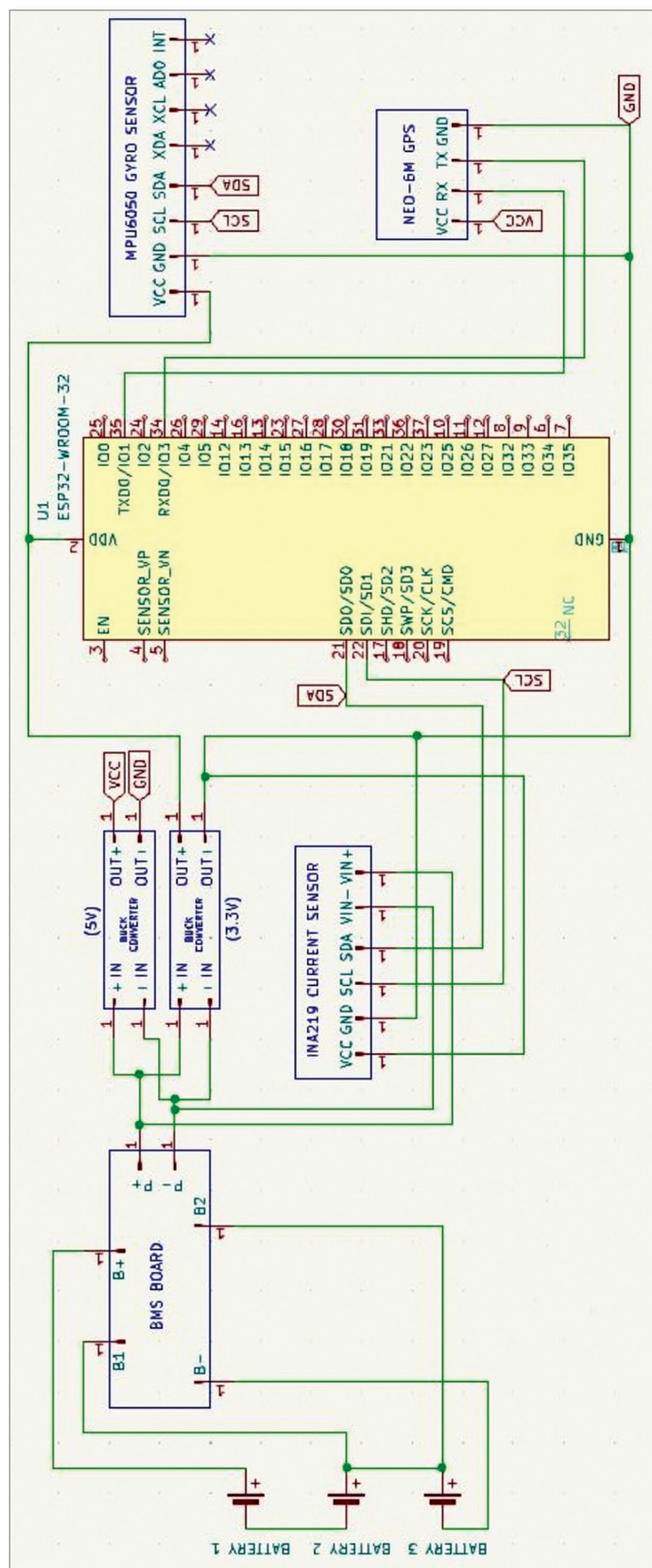


Fig. 1: Circuit diagram